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Volvo Group Trucks Technology

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T3 Platform

TSC Vision Cameras - Camera

1 DOCUMENT REVISION HISTORY 4

2 SCOPE AND PURPOSE OF DOCUMENT 4

3 TSC L1 4

3.1 Input Safety Goals 4
3.2 Input Justified Requirements 4
3.3 System design and requirements 4
3.3.1 Technical Design Vision Cameras Camera 4
3.3.1.1 Architecture elements 5
3.3.1.1.1 [Camera] Camera Module 5
3.3.1.1.2 [Camera] Image Processing 5
3.3.1.1.3 [Camera] Time 5
3.3.1.1.4 [Camera] Buffer 5
3.3.1.1.5 [Camera] Video Encoder 5
3.3.1.1.6 [Camera] Stream Generation / Transmission 5
3.3.1.1.7 [Vehicle] Receiver 5
3.3.2 First level technical safety requirements derived from safety goal SG - Incorrect Field of View 6
3.3.2.1 Justification 6
3.3.2.1.1 FTA Incorrect FoV (FSR6, FSR8, FSR9, FSR20) 7
3.3.3 First level technical safety requirements derived from safety goal SG - Stuck View 8
3.3.3.1 Justification 8
3.3.3.1.1 FTA Stuck Image (FSR3, FSR28, FSR9, FSR4) 9
3.3.4 First level technical safety requirements derived from safety goal SG - Delayed Image 10
3.3.4.1 Justification 10
3.3.4.1.1 FTA Delayed Image (FSR2, FSR34, FSR9, FSR35) 11
3.3.5 First level technical safety requirements derived from safety goal SG - Unclear Image 13
3.3.5.1 Justification 13
3.3.5.1.1 FTA Unclear View (FSR5, FSR39, FSR9, FSR40) 13
3.4 Technical safety requirements 15
3.4.1 TSR1 Nominal transmission time 15
3.4.2 TSR2 Transmission time 15
3.4.3 TSR3 Transmit video with integrity check 15
3.4.4 TSR4 Do not transmit unclear video 15
3.4.5 TSR5 Do not transmit wrong FoV 16
3.4.6 TSR6 FHTI 16
3.4.7 TSR7 Camera module safe state 16
3.4.8 TSR8 Camera module safe state recovery 17
3.4.9 TSR9 Receive and forward video 17
3.4.10 TSR10 Do not forward unclear video 17
3.4.11 TSR11 Do not forward wrong FoV 18
3.4.12 TSR12 FHTI ECU 18
3.4.13 TSR13 Camera ECU safe state 18
3.4.14 TSR14 Camera ECU safe state recovery 19
3.4.15 TSR15 Image timestamp insertion 19

3.4.16 TSR16 Synchronize time 19
3.4.17 TSR17 Received video transmission time 19
3.4.18 TSR18 Transmit video with timestamp 20
3.4.19 TSR19 Transmit Camera ID 20
4 ABBREVIATIONS 20
5 DEFINITIONS 20
6 REFERENCES 21
7 HISTORY AND CHANGELOG 21
7.1 Requirements 21
7.1.1 Active requirements list 21
7.1.2 New requirements 22
7.1.3 Updated requirements 22
7.1.4 Removed requirements 22

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1 Document Revision History

Rev	Date	Author	Description
0.0.1	2024-11-01	Jan Svensson	First draft
0.0.2	2025-08-18	Jan Svensson	Change of highest ASIL C to ASIL B Clarification of FHTI and Nominal latency Added Receiver element as clarification Added TSR19 Camera ID
0.0.3	2025-11-04	Arturo Feiteira	FTAs created.

2 Scope and purpose of document

The TSC for Vision Cameras - Camera covers the following:

- Technical Safety Requirements derived from Functional Safety Concept and Safety Goals
- Applicable Architecture is T3
- Diagnostic Features
- Concerning the parts of belonging to the camera.

Derived from FSC Vision Cameras

3 TSC L1

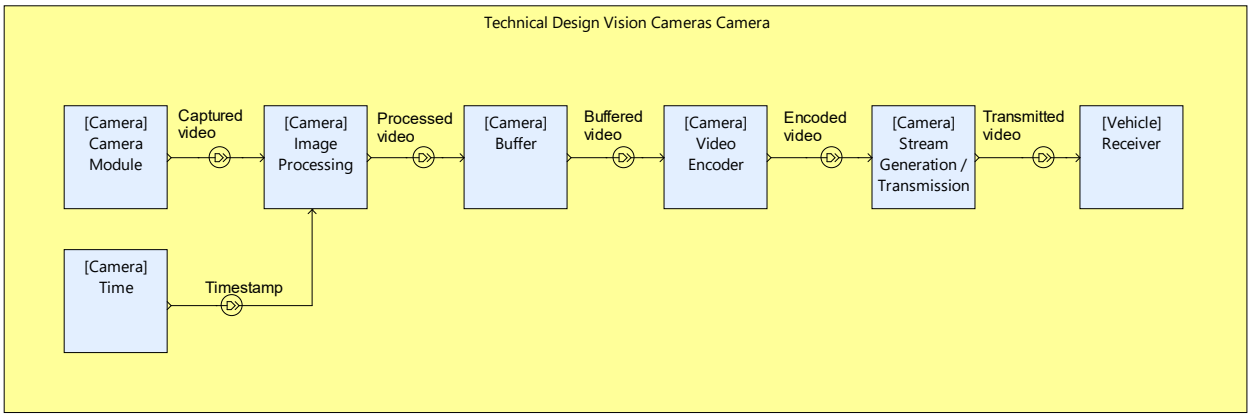
3.1 Input Safety Goals

3.2 Input Justified Requirements

3.3 System design and requirements

3.3.1 Technical Design Vision Cameras Camera

In the Technical Design the image pipeline is defined to consist of the elements Image Processing, Buffer, Video Encoder and Stream Generation/Transmission.



3.3.1.1 Architecture elements

3.3.1.1.1 [Camera] Camera Module

Camera module is responsible for image capture, converting incoming photons to a digital stream of pixels.

3.3.1.1.2 [Camera] Image Processing

Image processing element is responsible for any image processing that is applied before encoding video, like color & hue adjustment, fish eye restoration etc.

3.3.1.1.3 [Camera] Time

Time element is responsible for keeping a synchronized local time.

3.3.1.1.4 [Camera] Buffer

Buffer element represent the image buffer(s) in the image pipeline that could cause vulnerabilities due to storing old images. Ideally the buffer only stores small parts of the image before video encoding/compression.

3.3.1.1.5 [Camera] Video Encoder

Video Encoder element applies compression to the video stream before sending.

3.3.1.1.6 [Camera] Stream Generation / Transmission

Stream Generation/transmission element takes the compressed video stream and send to receiver.

3.3.1.1.7 [Vehicle] Receiver

Receiver element is the representation of receiver image pipeline elements and display element on TSC level, equals HMI Output element on FSC level.

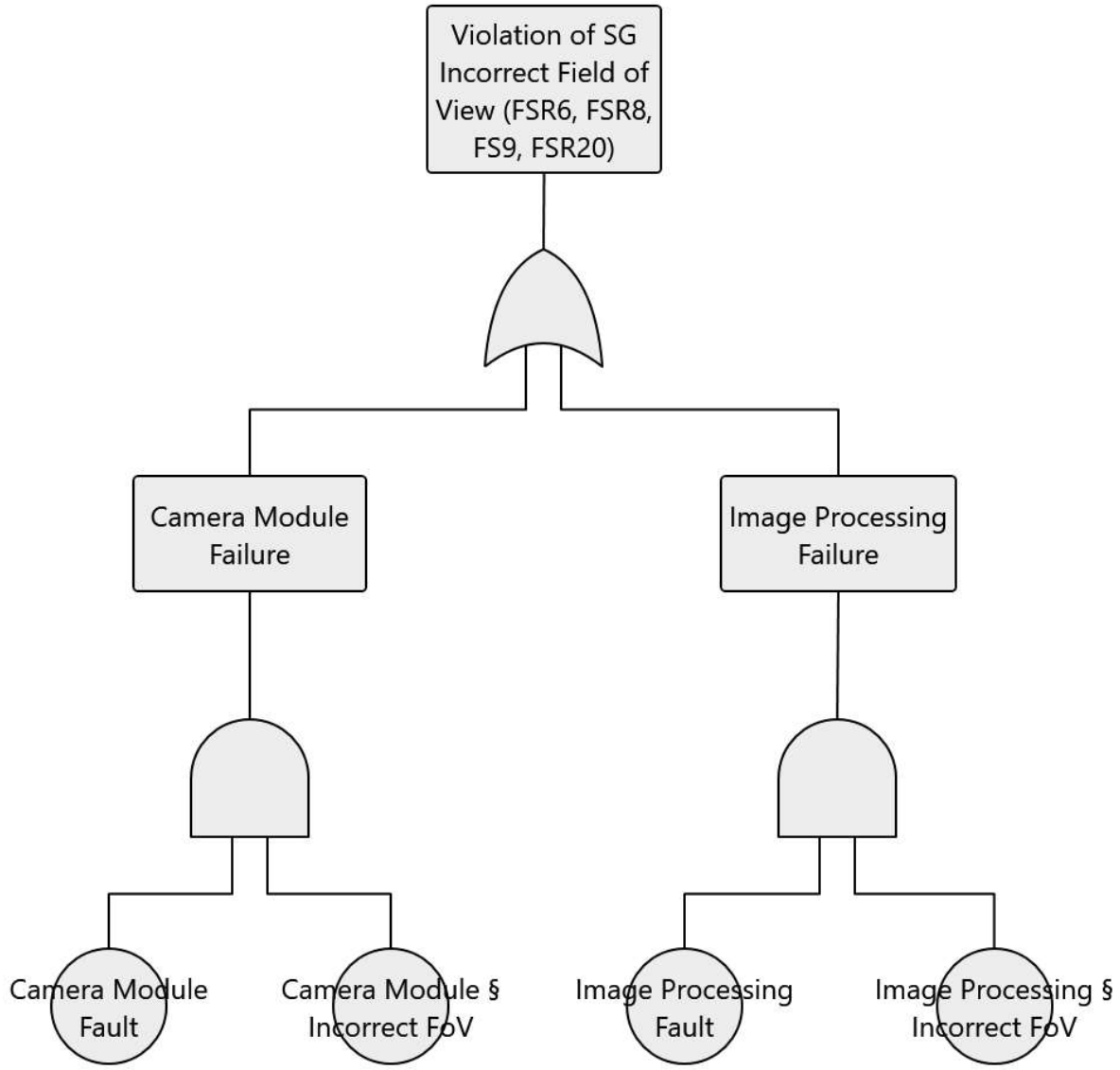
3.3.2 First level technical safety requirements derived from safety goal SG - Incorrect Field of View

3.3.2.1 Justification

Functional safety requirements above are derived based on SG Incorrect Field of View and following FSRs: FSR6, FSR8, FSR9, FSR20.

A fault tree analysis was carried out as schematically shown below. Requirements above were formulated based on that analysis.

3.3.2.1.1 FTA Incorrect FoV (FSR6, FSR8, FSR9, FSR20)



Textual Fault Tree for FTA Incorrect FoV (FSR6, FSR8, FSR9, FSR20)

Node	Node ASIL	Node Desc	Violated req	Violated Req ID	Violated Req ASIL	Violated Req SG ASIL	Faulty Component
Camera Module Failure							
Camera Module Fault							[Camera] Camera

Node	Node ASIL	Node Desc	Violated req	Violated Req ID	Violated Req ASIL	Violated Req SG ASIL	Faulty Component
							Module
Camera Module § Incorrect FoV			TSR5 Do not transmit wrong FoV	F_Req-38263	B		[Camera] Camera Module
Camera Module § Incorrect FoV			TSR6 FHTI	F_Req-38264	B		[Camera] Camera Module
Camera Module § Incorrect FoV			TSR7 Camera module safe state	F_Req-38265	B		[Camera] Camera Module
Image Processing Failure							
Image Processing Fault							[Camera] Image Processing
Image Processing § Incorrect FoV			TSR11 Do not forward wrong FoV	F_Req-38269	B		[Camera] Image Processing
Image Processing § Incorrect FoV			TSR12 FHTI ECU	F_Req-38291	B		[Camera] Image Processing
Image Processing § Incorrect FoV			TSR13 Camera ECU safe state	F_Req-38292	B		[Camera] Image Processing
Violation of SG Incorrect Field of View (FSR6, FSR8, FS9, FSR20)							

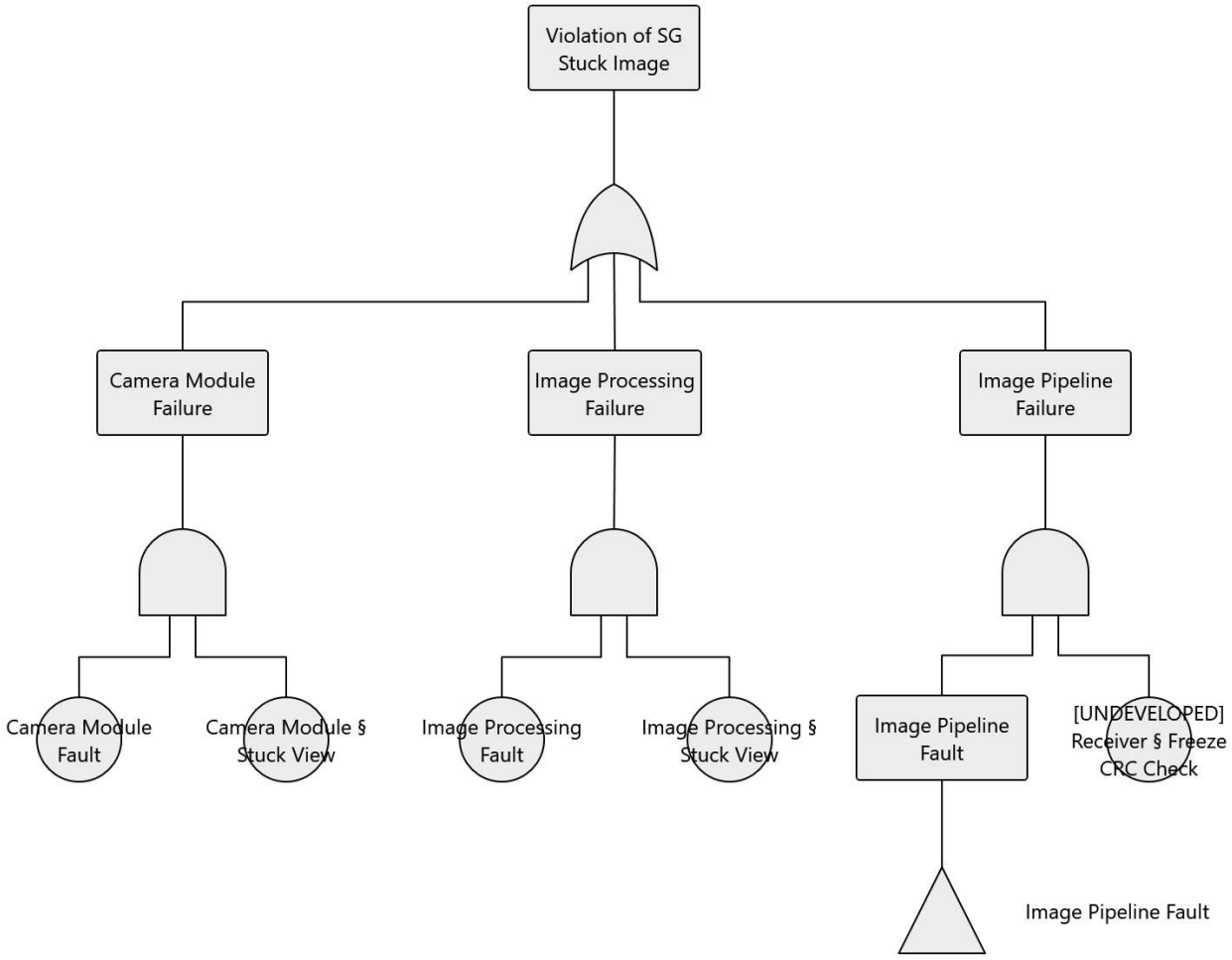
3.3.3 First level technical safety requirements derived from safety goal SG - Stuck View

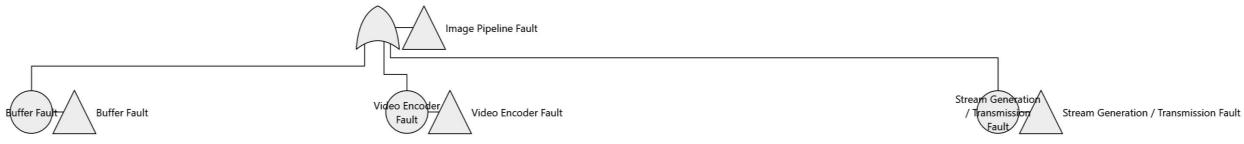
3.3.3.1 Justification

Functional safety requirements above are derived based on SG Incorrect Field of View and following FSRs: FSR3, FSR28, FSR9, FSR4.

A fault tree analysis was carried out as schematically shown below. Requirements above were formulated based on that analysis.

3.3.3.1.1 FTA Stuck Image (FSR3, FSR28, FSR9, FSR4)





Textual Fault Tree for FTA Stuck Image (FSR3, FSR28, FSR9, FSR4)

Node	Node ASIL	Node Desc	Violated req	Violated Req ID	Violated Req ASIL	Violated Req SG ASIL	Faulty Component
Buffer Fault							[Camera] Buffer
Camera Module Failure							
Camera Module							[Camera]

Node	Node ASIL	Node Desc	Violated req	Violated Req ID	Violated Req ASIL	Violated Req SG ASIL	Faulty Component
Fault							Camera Module
Camera Module § Stuck View			TSR3 Transmit video with integrity check	F_Req-38260	A		[Camera] Camera Module
Camera Module § Stuck View			TSR6 FHTI	F_Req-38264	B		[Camera] Camera Module
Camera Module § Stuck View			TSR7 Camera module safe state	F_Req-38265	B		[Camera] Camera Module
Image Pipeline Failure							
Image Pipeline Fault							
Image Processing Failure							
Image Processing Fault							[Camera] Image Processing
Image Processing § Stuck View			TSR12 FHTI ECU	F_Req-38291	B		[Camera] Image Processing
Image Processing § Stuck View			TSR13 Camera ECU safe state	F_Req-38292	B		[Camera] Image Processing
Image Processing § Stuck View			TSR9 Receive and forward video	F_Req-38267	B		[Camera] Image Processing
Stream Generation / Transmission Fault							[Camera] Stream Generation / Transmission
Video Encoder Fault							[Camera] Video Encoder
Violation of SG Stuck Image							
[UNDEVELOPED] Receiver § Freeze CRC Check							[Vehicle] Receiver

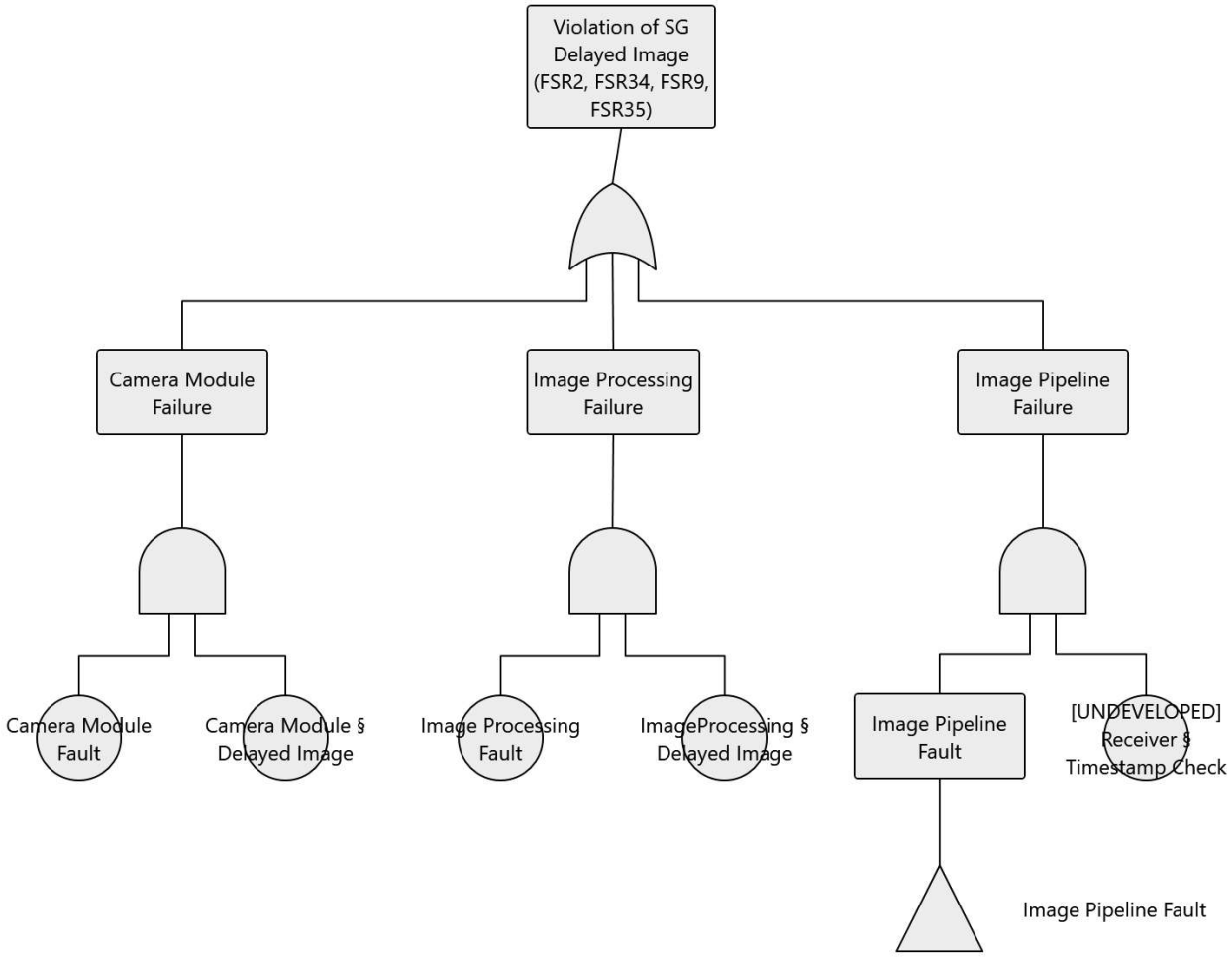
3.3.4 First level technical safety requirements derived from safety goal SG - Delayed Image

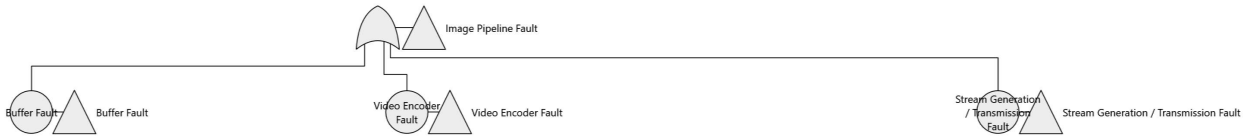
3.3.4.1 Justification

Functional safety requirements above are derived based on SG Incorrect Field of View and following FSRs: FSR2, FSR34, FSR9, FSR35.

A fault tree analysis was carried out as schematically shown below. Requirements above were formulated based on that analysis.

3.3.4.1.1 FTA Delayed Image (FSR2, FSR34, FSR9, FSR35)





Textual Fault Tree for FTA Delayed Image (FSR2, FSR34, FSR9, FSR35)

Node	Node ASIL	Node Desc	Violated req	Violated Req ID	Violated Req ASIL	Violated Req SG ASIL	Faulty Component
Buffer Fault							[Camera] Buffer
Camera Module							

Node	Node ASIL	Node Desc	Violated req	Violated Req ID	Violated Req ASIL	Violated Req SG ASIL	Faulty Component
Failure							
Camera Module Fault							[Camera] Camera Module
Camera Module § Delayed Image			TSR2 Transmission time	F_Req-38259	B		[Camera] Camera Module
Camera Module § Delayed Image			TSR6 FHTI	F_Req-38264	B		[Camera] Camera Module
Camera Module § Delayed Image			TSR7 Camera module safe state	F_Req-38265	B		[Camera] Camera Module
Image Pipeline Failure							
Image Pipeline Fault							
Image Processing Failure							
Image Processing Fault							[Camera] Image Processing
Image Processing § Delayed Image			TSR12 FHTI ECU	F_Req-38291	B		[Camera] Image Processing
Image Processing § Delayed Image			TSR13 Camera ECU safe state	F_Req-38292	B		[Camera] Image Processing
Image Processing § Delayed Image			TSR9 Receive and forward video	F_Req-38267	B		[Camera] Image Processing
Stream Generation / Transmission Fault							[Camera] Stream Generation / Transmission
Video Encoder Fault							[Camera] Video Encoder
Violation of SG Delayed Image (FSR2, FSR34, FSR9, FSR35)							
[UNDEVELOPED] Receiver § Timestamp Check							[Vehicle] Receiver

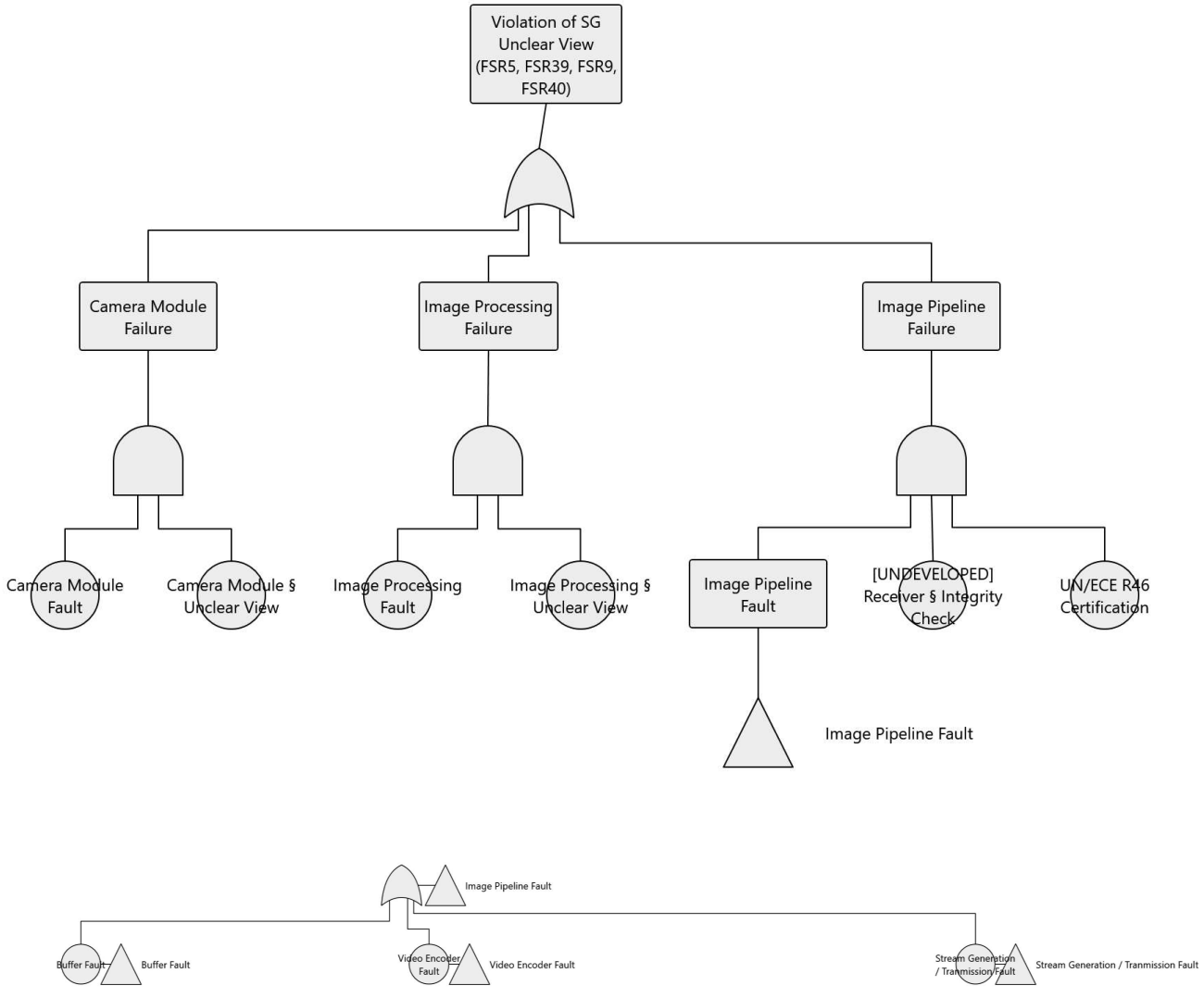
3.3.5 First level technical safety requirements derived from safety goal SG - Unclear Image

3.3.5.1 Justification

Functional safety requirements above are derived based on SG Unclear View and following FSRs: FSR6, FSR8, FSR9, FSR20.

A fault tree analysis was carried out as schematically shown below. Requirements above were formulated based on that analysis.

3.3.5.1.1 FTA Unclear View (FSR5, FSR39, FSR9, FSR40)



Textual Fault Tree for FTA Unclear View (FSR5, FSR39, FSR9, FSR40)

Node	Node ASIL	Node Desc	Violated req	Violated Req ID	Violated Req ASIL	Violated Req SG ASIL	Faulty Component
Buffer Fault							[Camera] Buffer
Camera Module Failure							
Camera Module Fault							[Camera] Camera Module
Camera Module § Unclear View			TSR4 Do not transmit unclear video	F_Req-38261	A		[Camera] Camera Module
Camera Module § Unclear View			TSR6 FHTI	F_Req-38264	B		[Camera] Camera Module
Camera Module § Unclear View			TSR7 Camera module safe state	F_Req-38265	B		[Camera] Camera Module
Image Pipeline Failure							
Image Pipeline Fault							
Image Processing Failure							
Image Processing Fault							[Camera] Image Processing
Image Processing § Unclear View			TSR10 Do not forward unclear video	F_Req-38268	A		[Camera] Image Processing
Image Processing § Unclear View			TSR12 FHTI ECU	F_Req-38291	B		[Camera] Image Processing
Image Processing § Unclear View			TSR13 Camera ECU safe state	F_Req-38292	B		[Camera] Image Processing
Stream Generation / Tranmission Fault							[Camera] Stream Generation / Transmission
UN/ECE R46 Certification							
Video Encoder Fault							[Camera] Video Encoder
Violation of SG Unclear View (FSR5, FSR39, FSR9, FSR40)							
[UNDEVELOPED] Receiver § Integrity Check			TSR3 Transmit video with integrity check	F_Req-38260	A		[Vehicle] Receiver

3.4 Technical safety requirements

The resulting technical safety requirements are listed below:

3.4.1 TSR1 Nominal transmission time

Requirement ID: F_Req-38258
Version: (1)
Status: CS Released
Changed: 2024-10-28 11:32

ASIL: **QM**()
*Allocated to element:***[Camera] Camera Module***, in Architecture:***Technical Design Vision Cameras Camera**

Video frames shall be transmitted within 50 ms from actual traffic situation.

3.4.2 TSR2 Transmission time

Requirement ID: F_Req-38259
Version: (2)
Status: CS Released
Changed: 2025-08-18 11:29

ASIL: **B**()
*Allocated to element:***[Camera] Camera Module***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Camera Module § Delayed Image***, in Fault Tree:***FTA Delayed Image (FSR2, FSR34, FSR9, FSR35)**

Video frames shall be transmitted within FHTI ms from actual traffic situation alternatively provide means to detect transmission delays by receiver.

3.4.3 TSR3 Transmit video with integrity check

Requirement ID: F_Req-38260
Version: (1)
Status: CS Released
Changed: 2024-10-28 11:34

ASIL: **A**()
*Allocated to element:***[Camera] Camera Module***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Camera Module § Stuck View***, in Fault Tree:***FTA Stuck Image (FSR3, FSR28, FSR9, FSR4)**
*Mentioned as violated requirement in Fault Event:***[UNDEVELOPED] Receiver § Integrity Check***, in Fault Tree:***FTA Unclear View (FSR5, FSR39, FSR9, FSR40)**

Video frames shall be transmitted such that it is possible to detect transmission integrity by receiver.

3.4.4 TSR4 Do not transmit unclear video

Requirement ID: F_Req-38261
Version: (1)
Status: CS Released
Changed: 2024-11-01 09:16

ASIL: **A**()
*Allocated to element:***[Camera] Camera Module***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Camera Module § Unclear View***, in Fault Tree:***FTA Unclear View (FSR5, FSR39, FSR9, FSR40)**

Video frames causing an unclear view (too dim, too low contrast, too much disturbances or noise, wrong resolution, flicker, too low framerate) shall not be transmitted.
Fulfilling this requirement means performing the activities needed to fulfil UN/ECE R46 from a camera module point of view.

3.4.5 TSR5 Do not transmit wrong FoV

Requirement ID: F_Req-38263
Version: (1)
Status: CS Released
Changed: 2024-10-28 12:08

ASIL: **B**()
*Allocated to element:***[Camera] Camera Module***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Camera Module § Incorrect FoV***, in Fault Tree:***FTA Incorrect FoV (FSR6, FSR8, FSR9, FSR20)**

The Camera module shall not transmit video frames with wrong field of view (e.g. cropped, rotated, flipped or partial images) unless requested by receiver.

3.4.6 TSR6 FHTI

Requirement ID: F_Req-38264
Version: (2)
Status: CS Released
Changed: 2025-08-18 11:27

ASIL: **B**()
*Allocated to element:***[Camera] Camera Module***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Camera Module § Incorrect FoV***, in Fault Tree:***FTA Incorrect FoV (FSR6, FSR8, FSR9, FSR20)**
*Mentioned as violated requirement in Fault Event:***Camera Module § Stuck View***, in Fault Tree:***FTA Stuck Image (FSR3, FSR28, FSR9, FSR4)**
*Mentioned as violated requirement in Fault Event:***Camera Module § Delayed Image***, in Fault Tree:***FTA Delayed Image (FSR2, FSR34, FSR9, FSR35)**
*Mentioned as violated requirement in Fault Event:***Camera Module § Unclear View***, in Fault Tree:***FTA Unclear View (FSR5, FSR39, FSR9, FSR40)**

Fault Handling Time Interval (FHTI) shall be 400 ms.
Note: FHTI shared with ECU (eg 200 + 200 ms) unless failures are proven independent from ECU.

3.4.7 TSR7 Camera module safe state

Requirement ID: F_Req-38265
Version: (2)
Status: CS Released
Changed: 2025-08-18 11:25

ASIL: **B**()
*Allocated to element:***[Camera] Camera Module***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Camera Module § Incorrect FoV***, in Fault Tree:***FTA Incorrect FoV**

(FSR6, FSR8, FSR9, FSR20)
*Mentioned as violated requirement in Fault Event:***Camera Module § Stuck View***, in Fault Tree:***FTA Stuck Image (FSR3, FSR28, FSR9, FSR4)**
*Mentioned as violated requirement in Fault Event:***Camera Module § Delayed Image***, in Fault Tree:***FTA Delayed Image (FSR2, FSR34, FSR9, FSR35)**
*Mentioned as violated requirement in Fault Event:***Camera Module § Unclear View***, in Fault Tree:***FTA Unclear View (FSR5, FSR39, FSR9, FSR40)**

Failure to correctly transmit video frames shall trigger a transition to Camera module safe state, defined as halted image transmission and/or a notification to the receiver.

3.4.8 TSR8 Camera module safe state recovery

Requirement ID: F_Req-38266
Version: (1)
Status: CS Released
Changed: 2024-10-31 09:32

ASIL: **QM**()
*Allocated to element:***[Camera] Camera Module***, in Architecture:***Technical Design Vision Cameras Camera**

In Camera module safe state, the camera shall monitor whether recovery from safe state can occur. If so, the camera shall transition back to normal operation.

3.4.9 TSR9 Receive and forward video

Requirement ID: F_Req-38267
Version: (2)
Status: CS Released
Changed: 2025-08-18 11:21

ASIL: **B**()
*Allocated to element:***[Camera] Image Processing***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Image Processing § Stuck View***, in Fault Tree:***FTA Stuck Image (FSR3, FSR28, FSR9, FSR4)**
*Mentioned as violated requirement in Fault Event:***Image Processing § Delayed Image***, in Fault Tree:***FTA Delayed Image (FSR2, FSR34, FSR9, FSR35)**

Video frames received by image processing in Image pipeline shall be correctly received, processed and forwarded to Receiver within FHTI ms from actual traffic situation, else go to Camera ECU safe state.

3.4.10 TSR10 Do not forward unclear video

Requirement ID: F_Req-38268
Version: (1)
Status: CS Released
Changed: 2024-11-01 09:16

ASIL: **A**()
*Allocated to element:***[Camera] Image Processing***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Image Processing § Unclear View***, in Fault Tree:***FTA Unclear View (FSR5, FSR39, FSR9, FSR40)**

Video frames causing an unclear view (too dim, too low contrast, too much disturbances or noise, wrong resolution, flicker, too low framerate) shall not be forwarded.
Fulfilling this requirement means performing the activities needed to fulfil UN/ECE R46 from a camera image

processing point of view.

3.4.11 TSR11 Do not forward wrong FoV

Requirement ID: F_Req-38269
Version: (1)
Status: CS Released
Changed: 2024-10-31 09:33

ASIL: **B**()
*Allocated to element:***[Camera] Image Processing***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Image Processing § Incorrect FoV***, in Fault Tree:***FTA Incorrect FoV (FSR6, FSR8, FSR9, FSR20)**

The Camera image processing shall not forward video frames with wrong field of view (eg cropped, rotated, flipped or partial images not requested by receiver), else go to Camera ECU safe state.

3.4.12 TSR12 FHTI ECU

Requirement ID: F_Req-38291
Version: (2)
Status: CS Released
Changed: 2025-08-18 11:22

ASIL: **B**()
*Allocated to element:***[Camera] Image Processing***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Image Processing § Incorrect FoV***, in Fault Tree:***FTA Incorrect FoV (FSR6, FSR8, FSR9, FSR20)**
*Mentioned as violated requirement in Fault Event:***Image Processing § Stuck View***, in Fault Tree:***FTA Stuck Image (FSR3, FSR28, FSR9, FSR4)**
*Mentioned as violated requirement in Fault Event:***Image Processing § Delayed Image***, in Fault Tree:***FTA Delayed Image (FSR2, FSR34, FSR9, FSR35)**
*Mentioned as violated requirement in Fault Event:***Image Processing § Unclear View***, in Fault Tree:***FTA Unclear View (FSR5, FSR39, FSR9, FSR40)**

Fault Handling Time Interval (FHTI) shall be 400 ms.
Note: FHTI shared with Camera module (eg 200 + 200 ms) unless failures are proven independent from Camera module.

3.4.13 TSR13 Camera ECU safe state

Requirement ID: F_Req-38292
Version: (2)
Status: CS Released
Changed: 2025-08-18 11:23

ASIL: **B**()
*Allocated to element:***[Camera] Image Processing***, in Architecture:***Technical Design Vision Cameras Camera**
*Mentioned as violated requirement in Fault Event:***Image Processing § Incorrect FoV***, in Fault Tree:***FTA Incorrect FoV (FSR6, FSR8, FSR9, FSR20)**
*Mentioned as violated requirement in Fault Event:***Image Processing § Stuck View***, in Fault Tree:***FTA Stuck Image (FSR3, FSR28, FSR9, FSR4)**
*Mentioned as violated requirement in Fault Event:***Image Processing § Delayed Image***, in Fault Tree:***FTA Delayed Image (FSR2, FSR34, FSR9, FSR35)**
*Mentioned as violated requirement in Fault Event:***Image Processing § Unclear View***, in Fault Tree:***FTA Unclear View (FSR5, FSR39, FSR9, FSR40)**

Failure to correctly forward video frames shall trigger a transition to Camera ECU safe state, defined as forwarding of blacked out images and/or a notification to the receiver.
Note: Safe state transitions are not mandatory as design choices that prevent the need to supervise could be applied.

3.4.14 TSR14 Camera ECU safe state recovery

Requirement ID: F_Req-38293
Version: (1)
Status: CS Released
Changed: 2024-10-31 09:34

ASIL: **QM**()
*Allocated to element:***[Camera] Image Processing***, in Architecture:***Technical Design Vision Cameras Camera**

In Camera ECU safe state, the camera shall monitor whether recovery from safe state can occur. If so, the camera shall transition back to normal operation.

3.4.15 TSR15 Image timestamp insertion

Requirement ID: F_Req-38294
Version: (1)
Status: CS Released
Changed: 2024-10-29 12:41

ASIL: **QM**()
*Allocated to element:***[Camera] Time***, in Architecture:***Technical Design Vision Cameras Camera**

A local time shall be used to generate an image timestamp to be inserted into each image.

3.4.16 TSR16 Synchronize time

Requirement ID: F_Req-38298
Version: (1)
Status: CS Released
Changed: 2024-10-29 12:42

ASIL: **QM**()
*Allocated to element:***[Camera] Time***, in Architecture:***Technical Design Vision Cameras Camera**

A local time shall be synchronized with receiver time.

3.4.17 TSR17 Received video transmission time

Requirement ID: F_Req-38299
Version: (1)
Status: CS Released
Changed: 2024-10-29 12:44

ASIL: **QM**()
*Allocated to element:***[Camera] Buffer***, in Architecture:***Technical Design Vision Cameras Camera**
*Allocated to element:***[Camera] Video Encoder***, in Architecture:***Technical Design Vision Cameras Camera**
*Allocated to element:***[Camera] Stream Generation / Transmission***, in Architecture:***Technical Design Vision Cameras Camera**

Video frames received by Image pipeline shall be transmitted to the receiver within 50 ms from reception.

3.4.18 TSR18 Transmit video with timestamp

Requirement ID: F_Req-38300
Version: (1)
Status: CS Released
Changed: 2024-10-29 12:45

ASIL: **QM**()
*Allocated to element:***[Camera] Stream Generation / Transmission***, in Architecture:***Technical Design Vision Cameras Camera**

Video frames shall be transmitted including an image timestamp such that it is possible to detect image delays by receiver.

3.4.19 TSR19 Transmit Camera ID

Requirement ID: F_Req-45283
Version: (1)
Status: CS Released
Changed: 2025-08-19 12:28

ASIL: **QM**()
*Allocated to element:***[Camera] Stream Generation / Transmission***, in Architecture:***Technical Design Vision Cameras Camera**

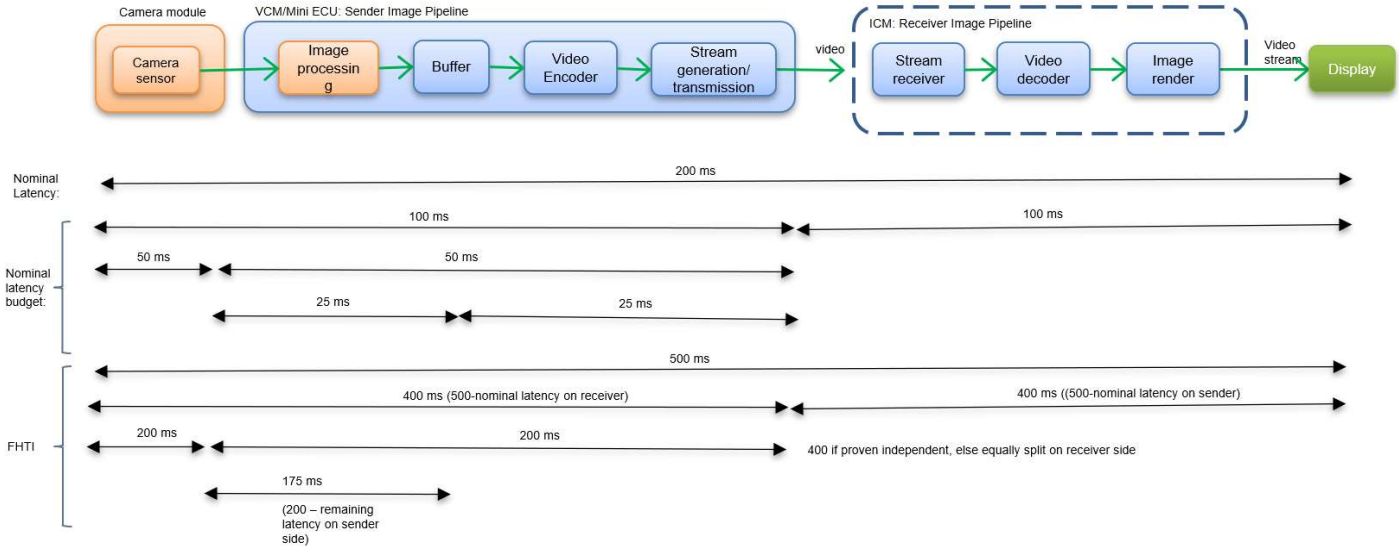
A camera ID originating from Receiver shall be embedded in image such that it is possible to detect a correct video stream among several streams by receiver.

4 Abbreviations

Abbreviation	Meaning	Description
ECE	Economic Commission for Europe	
ECU	Electronic Control Unit	
FHTI	Fault Handling Time Interval	Sum of fault detection time interval and the fault reaction time interval. Need to be < FTTI.
UN	United Nations	

5 Definitions

FHTI and Nominal latency breakdown: Clarification to requriements on FHTI and Nominal (QM) latency figures. The breakdown shows budgetary figures, meaning that they can be negotioated between elements as long as total FHTI and nominal latency is kept.



6 References

Reference	Document Number	Description	Referenced Document Version
UN/ECE R46		UN/ECE R46 regulation, https://unece.org/fileadmin/DAM/trans/main/wp29/wp29regs/2016/R046r6e.pdf	(1)
FSC Vision Cameras	50466650		(1)

7 History and ChangeLog

7.1 Requirements

7.1.1 Active requirements list

Requirement ID	Name	Version	Status
F_Req-38258	TSR1 Nominal transmission time	(1)	CS Released
F_Req-38259	TSR2 Transmission time	(2)	CS Released
F_Req-38260	TSR3 Transmit video with integrity check	(1)	CS Released
F_Req-38261	TSR4 Do not transmit unclear video	(1)	CS Released
F_Req-38263	TSR5 Do not transmit wrong FoV	(1)	CS Released
F_Req-38264	TSR6 FHTI	(2)	CS Released
F_Req-38265	TSR7 Camera module safe state	(2)	CS Released
F_Req-38266	TSR8 Camera module safe state recovery	(1)	CS

Requirement ID	Name	Version	Status
			Released
F_Req-38267	TSR9 Receive and forward video	(2)	CS Released
F_Req-38268	TSR10 Do not forward unclear video	(1)	CS Released
F_Req-38269	TSR11 Do not forward wrong FoV	(1)	CS Released
F_Req-38291	TSR12 FHTI ECU	(2)	CS Released
F_Req-38292	TSR13 Camera ECU safe state	(2)	CS Released
F_Req-38293	TSR14 Camera ECU safe state recovery	(1)	CS Released
F_Req-38294	TSR15 Image timestamp insertion	(1)	CS Released
F_Req-38298	TSR16 Synchronize time	(1)	CS Released
F_Req-38299	TSR17 Received video transmission time	(1)	CS Released
F_Req-38300	TSR18 Transmit video with timestamp	(1)	CS Released
F_Req-45283	TSR19 Transmit Camera ID	(1)	CS Released

7.1.2 New requirements

7.1.3 Updated requirements

7.1.4 Removed requirements